Bayesian inference: Basics and application for an engineering structure

Ritesh GUPTA

Franck BOURRIER

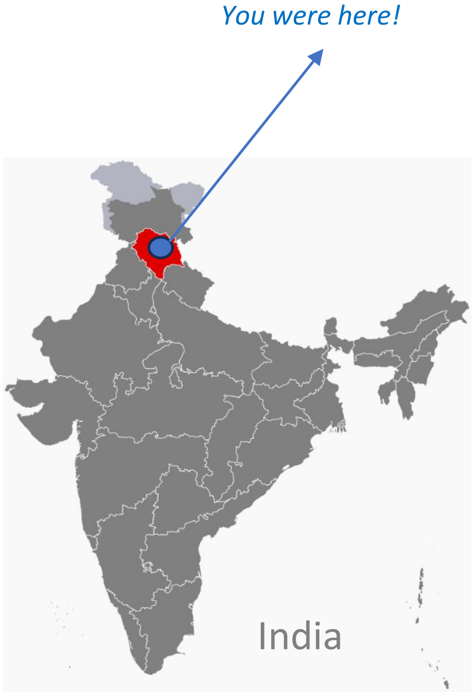
Vincent ACARY

Stéphane LAMBERT

Ritesh GUPTA

Believe in yourself, you are what you are... Gratitude reciprocates.

Personal Website

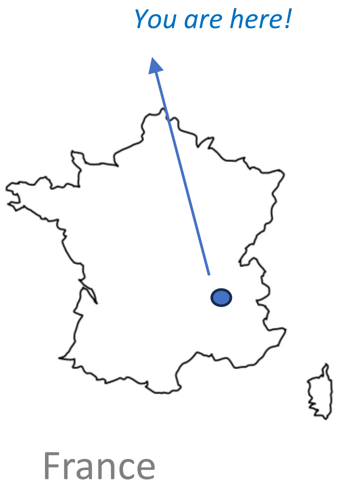


Kullu

(Hometown: 1993-2010)

Grenoble

(Second Hometown: 2015 – present)
Long-term residency holder



Bayesian inference basics and application

What is done? Why is done? How is done?

Non-smooth contact dynamics (NSCD) based model of an articulated concrete blocks structure subjected to rock fall impact : definition , calibration and inverse analysis

Non-smooth contact dynamics (NSCD)

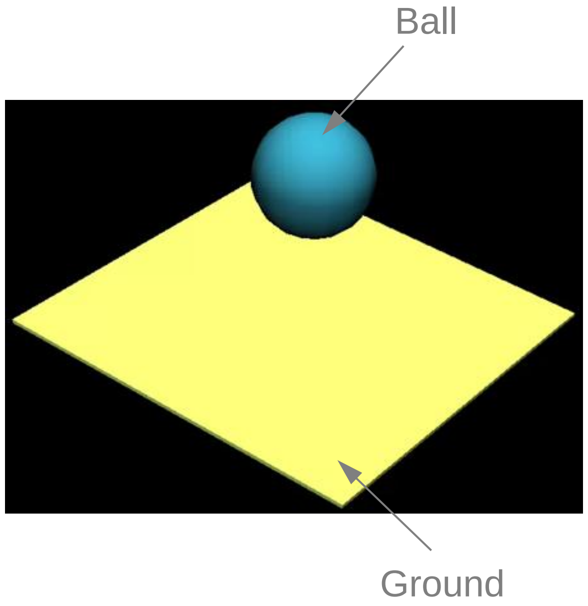
Dynamic systems with non-smoothness of their time evolution and their constitutive equations.

❖ **Siconos:** software to model NSCD systems. Three components:

- ❑ Numerics
- ❑ Kernel
- ❑ Front-end

Python interface

Demonstration

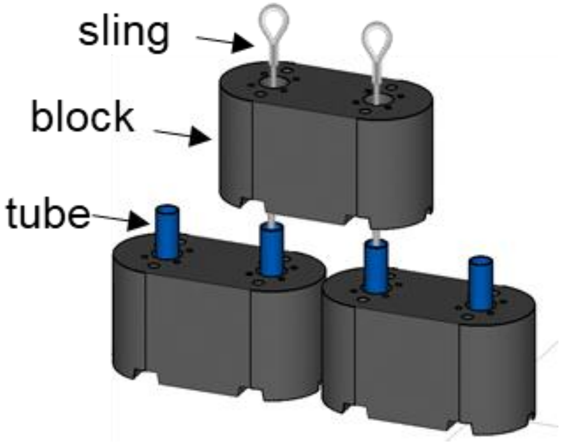


❖ **Front-end features:** easy-access tools for users comprising:

- ❑ *System definition*
  - ✓ Geometrical parameters
  - ✓ Mechanical characteristics
- ❑ *Interaction law*
  - ✓ Restitution coefficient (e)
  - ✓ Friction coefficient (μ)
- ❑ *Objects*
  - ✓ Primitive & user-defined shapes as contactors
  - ✓ Global location and orientation

Articulated concrete blocks structure

Newly proposed design of staggered concrete blocks (Bloc Armé®) as a protection structure



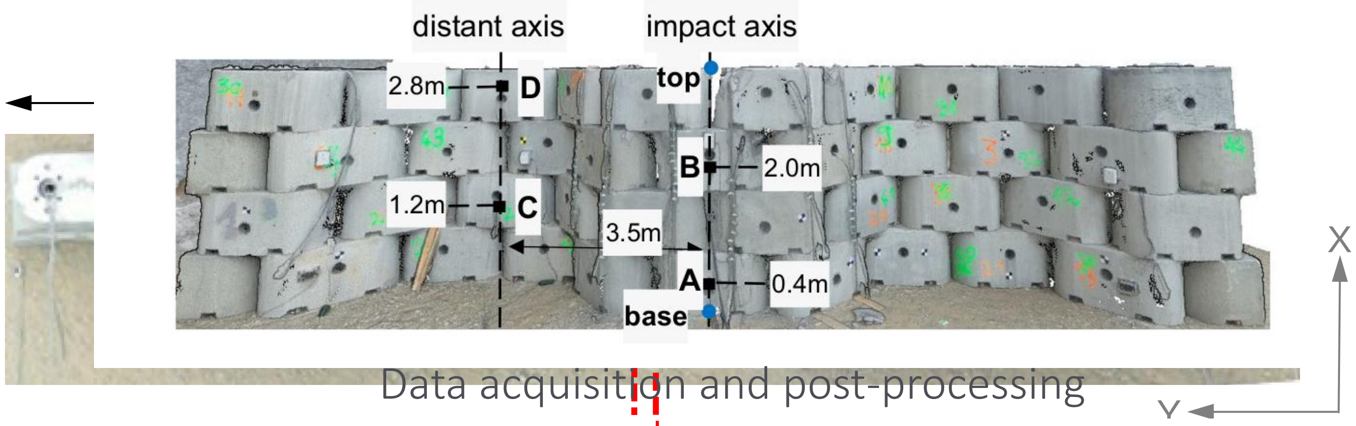
Block connection concept

GÉOLITHE

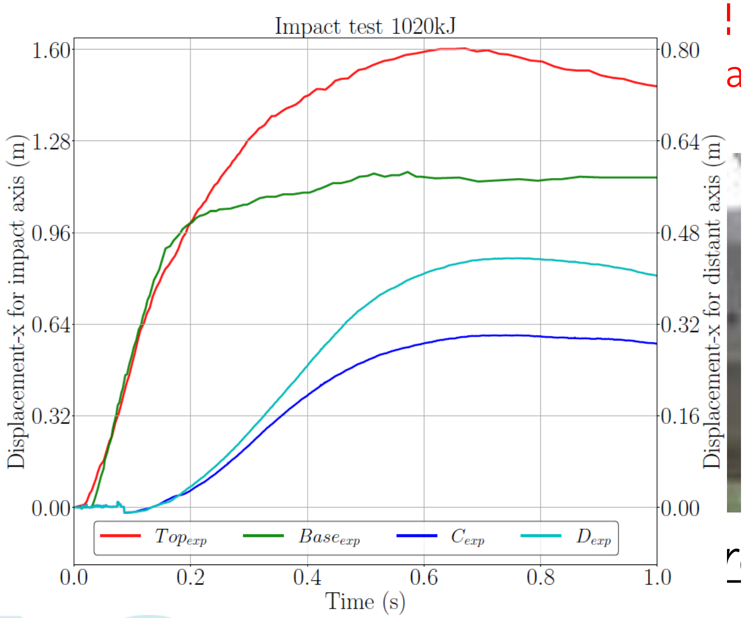
Rock fall impact

on-site impact tests using ETAG® projectile and data acquisition

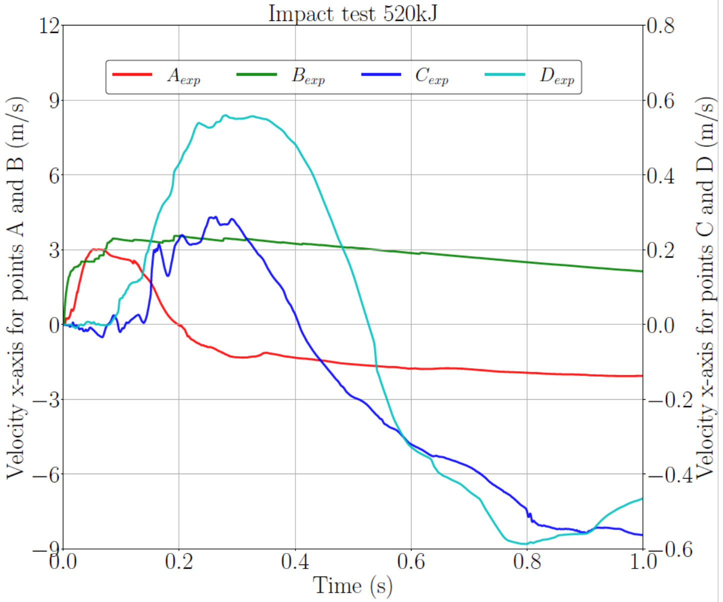
Destroyed in a second



Data acquisition and post-processing



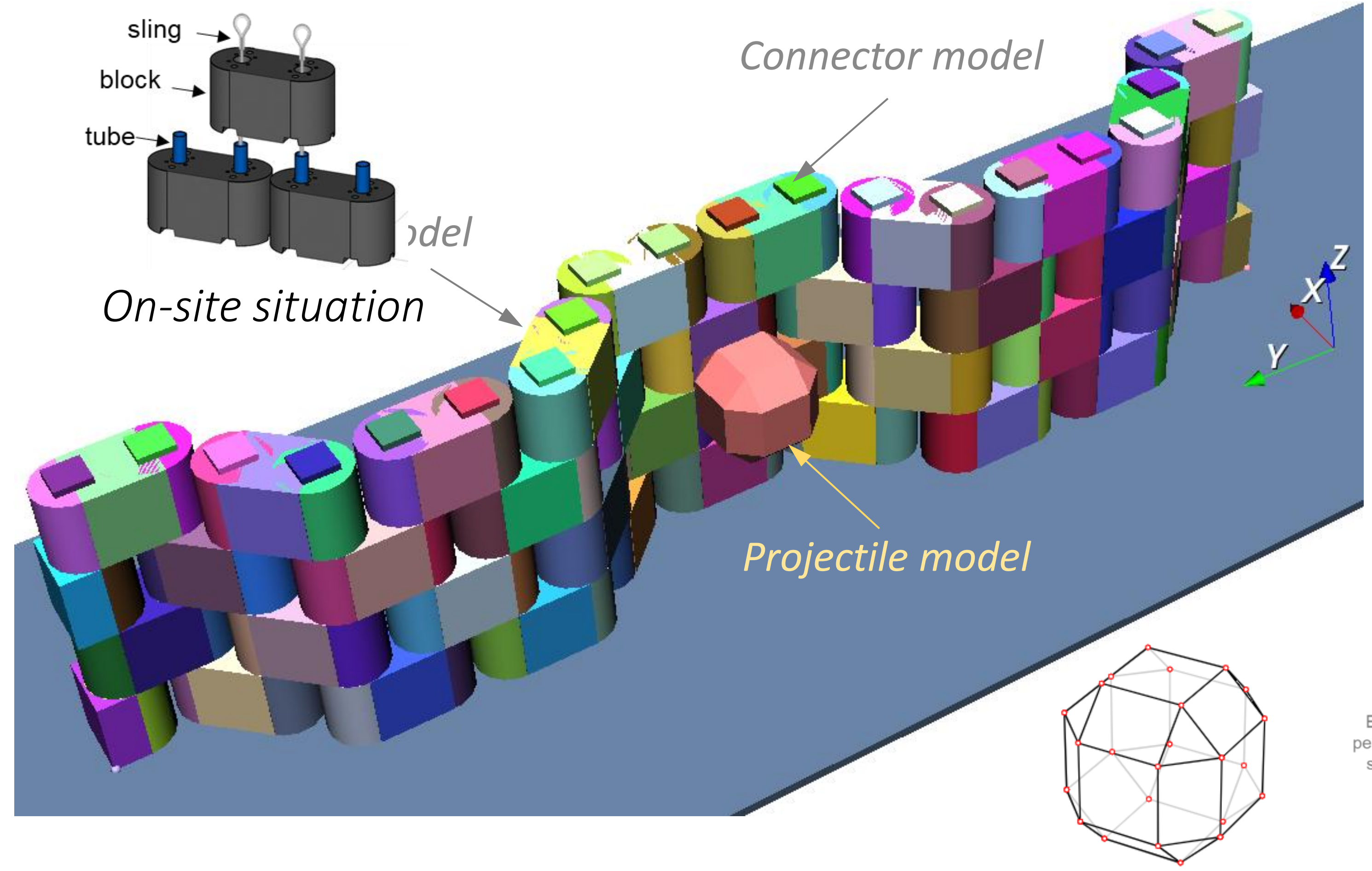
Displacement
(along x-axis)



Velocity
(as integrated acceleration)

Model definition

The NSCD based model of the articulated structure, inclusive of its features: modelling in SICONOS



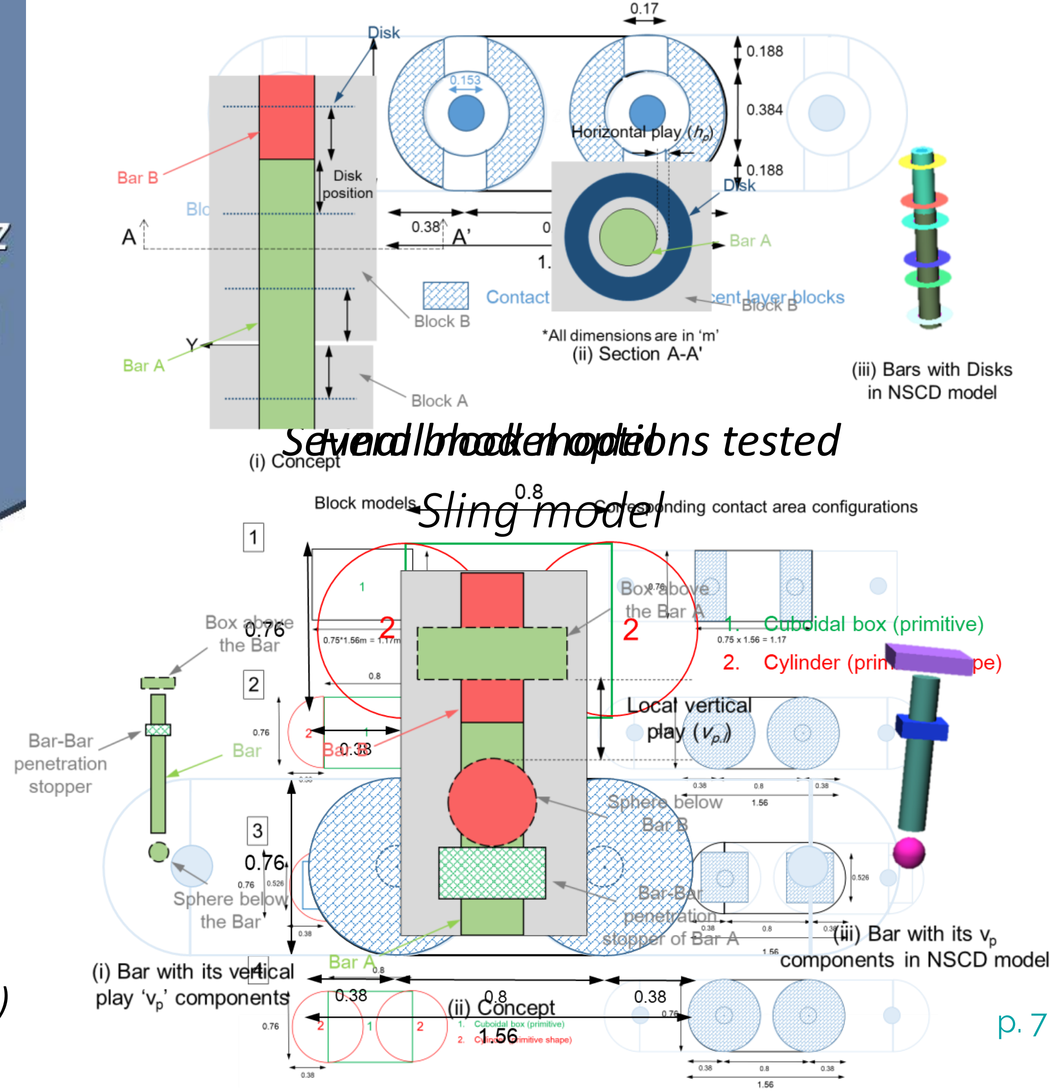
2D surface mesh
(using Convex Hull feature)

Tube model and interaction with block

On-site situation

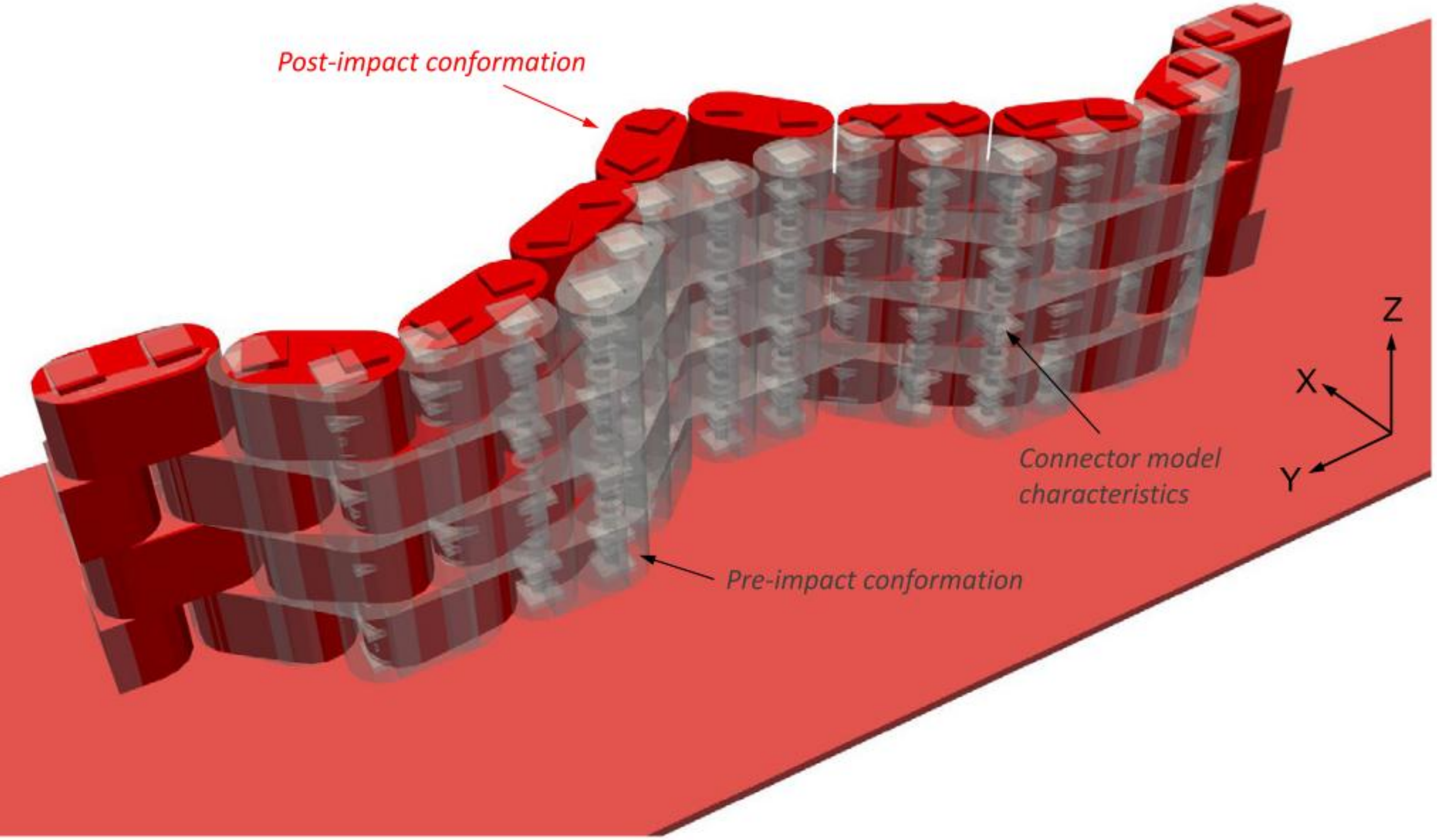
Several block models tested

Sling model

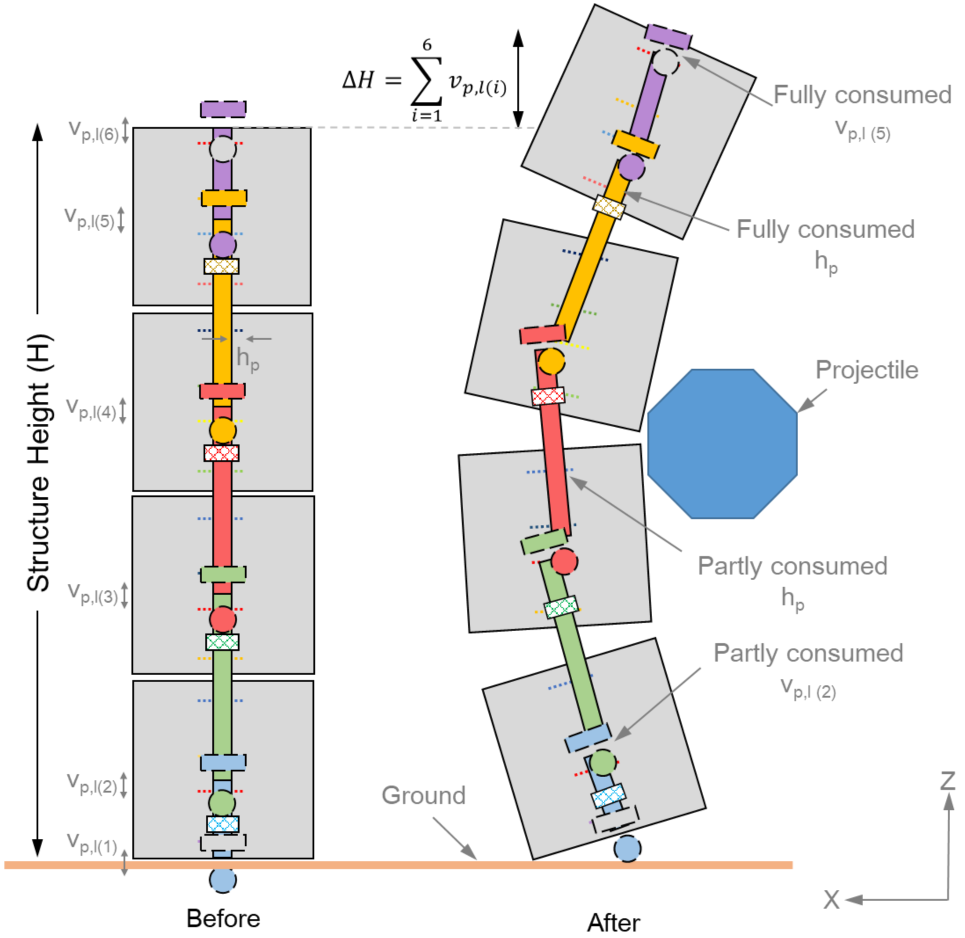


Model run

Digital analogues of on-site structure



Simulation demonstration – Computation time 20-25 mins

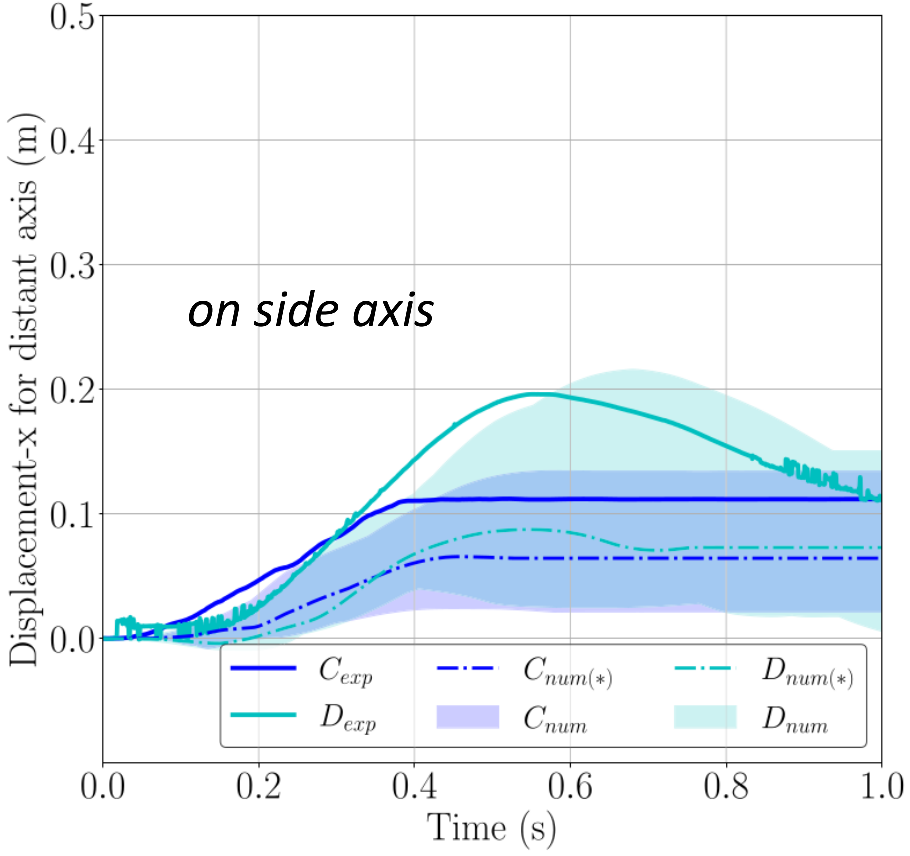


Connector model in action - conceptual

Model calibration

Purpose, strategy and success

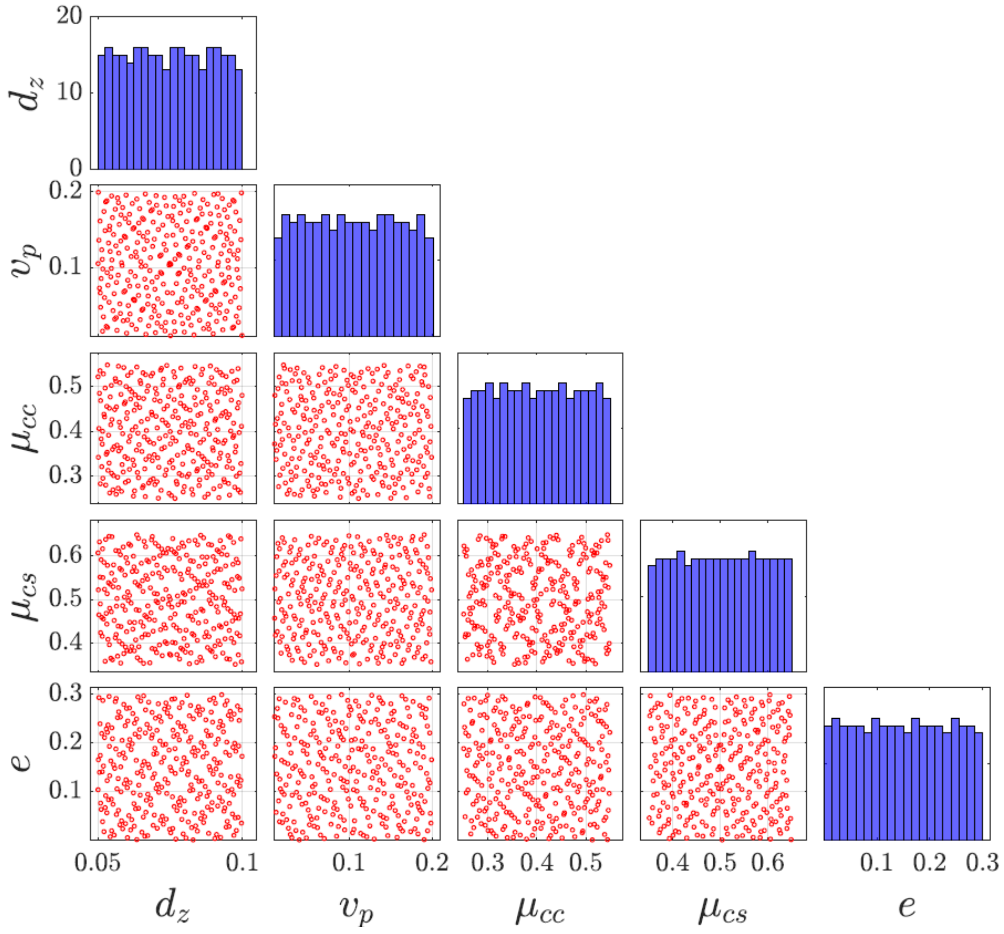
Which combination reproduces experimental observations?



A ribbon of possible outcomes

Model parameters

Parameter	Possible range	Mean value	Unit
Bar-block contactor disk position (d_z)	5 - 10	7.5	cm
Vertical play (v_p)	1 - 20	9.5	cm
Friction coefficient concrete-concrete (μ_{cc})	0.25 - 0.55	0.4	(-)
Friction coefficient concrete-soil (μ_{cs})	0.35 - 0.65	0.5	(-)
Restitution coefficient (e)	0 - 0.3	0.15	(-)



Statistical creation of input sample of size 300

Model calibration

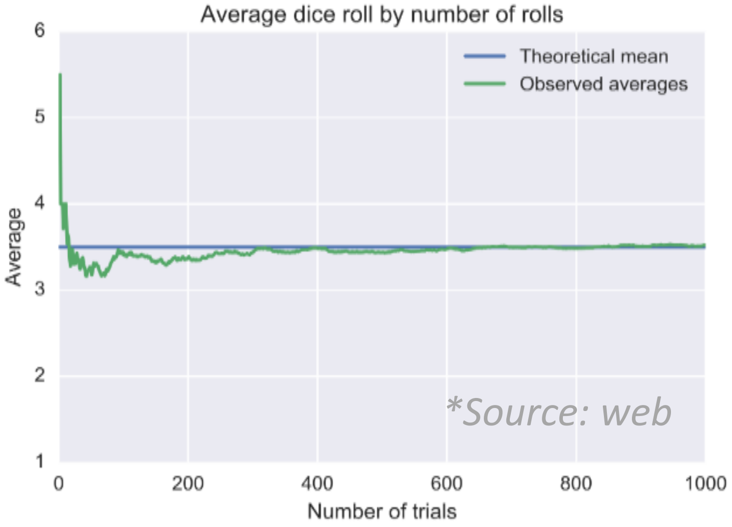
Purpose, strategy and success

NSCD model
(non-calibrated)

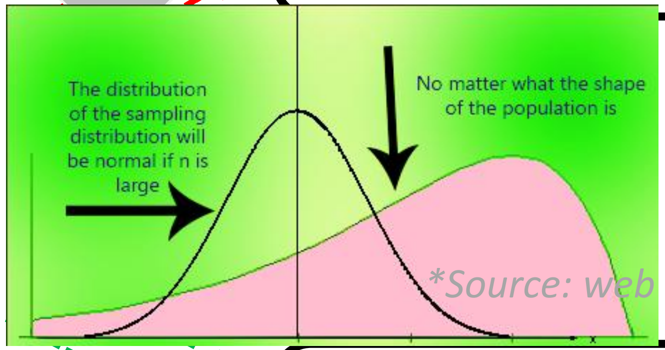
❖ Stochastic analysis … Why?
- *law of large numbers*
- *central limit theorem*

Surrogate of NSCD model
(non-calibrated)

Illustration



Law of large numbers



Central limit theorem

NSCD model
(calibrated)

Surrogate NSCD
model (calibrated)

Computationally expensive (~25 mins per simulation) and thereby impractical route

Polynomial Chaos Expansion (PCE) based metamodeling technique

Negligible computational cost (microseconds) and hence highly practical route

Stochastic concepts

Overview

- PCE based metamodel
  - *Equivalent mathematical relation*
  - *Like a linear regression*
  - *Accurate enough?*
- Bayesian inference
  - *Application of Bayes' theorem*
  - *Representation of changing beliefs*
  - *Two components:*
    - *Hypothesis (H)*
    - *Evidence (E)*
- So what?
  - *A particular set of input parameters (H) is the best fit with the experimental data (E).*

Illustration

Bayes' theorem

Posterior

Likelihood

Prior

$$P(H \mid E) = \frac{P(E \mid H) \cdot P(H)}{P(E)}$$

Probability of Evidence

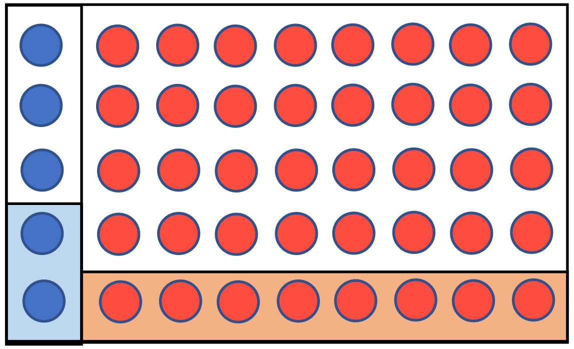
Practical example

A person likes climbing. | *E*

Is he Grenoblois or Parisien??

Grenoblois like climbing. | *H*

11.11% probability of being Grenoblois.



$$P(H) = \frac{5}{45} = \frac{1}{9} = 11.11\% \qquad P(E|H) = \frac{2}{5} = 40\%$$

$$P(E) = \frac{10}{45} = 22.22\% \qquad P(H|E) = 20\%$$

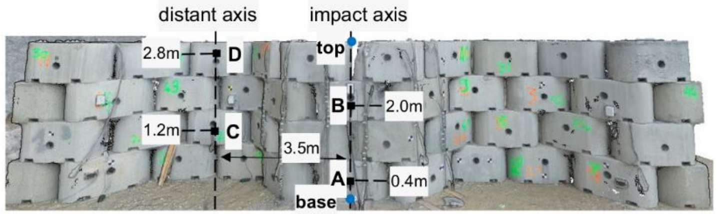
20% probability of being Grenoblois.

Les Grenoblois (5) *Les Parisiens (40)*

**Demo idea credits: 3Blue1Brown YouTube channel*

PCE based metamodel of NSCD model

Let’s do each computation in microseconds



- Abundant database: 300 displacement data-points recorded at 400 time instances

- Three metamodels for each data acquisition location

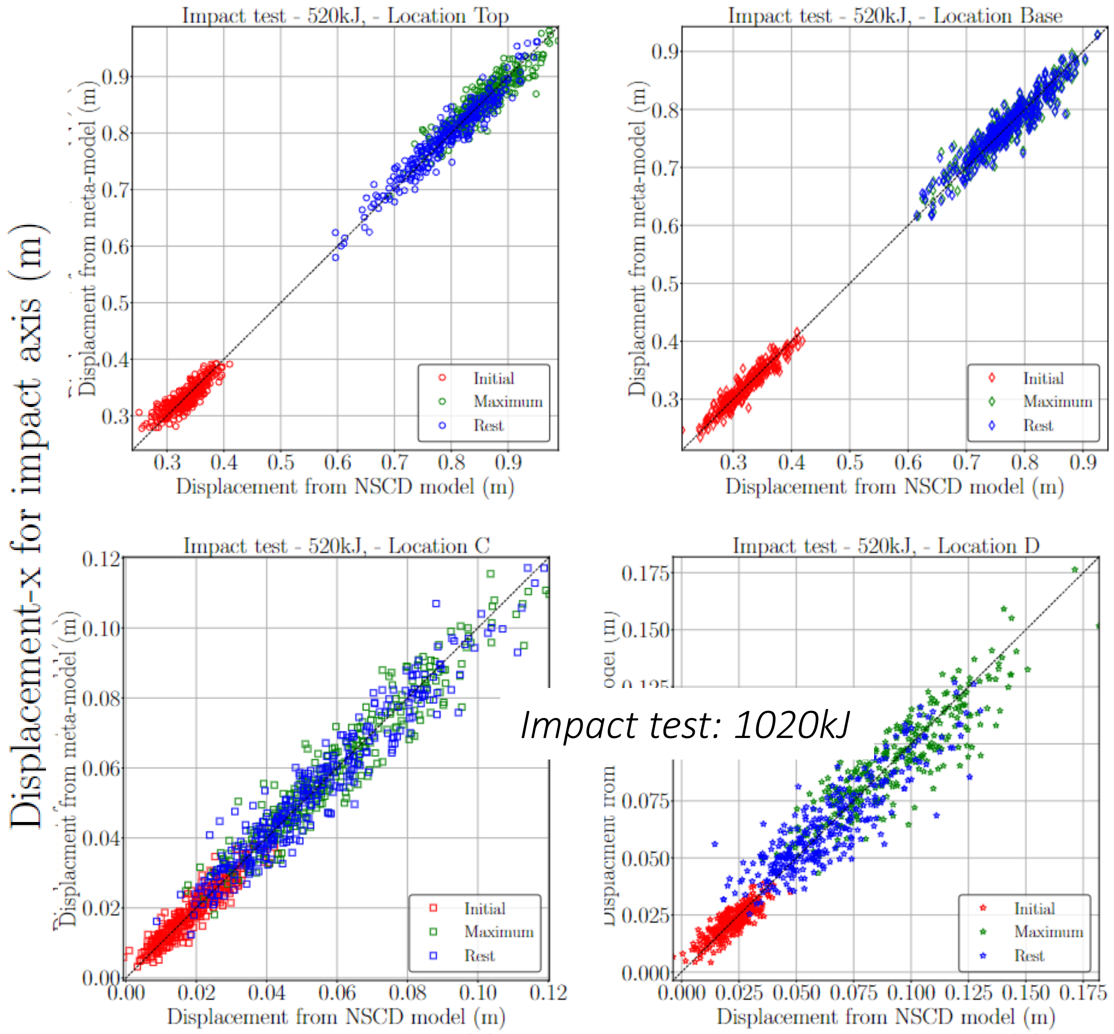
- How accurate is your metamodel?
  - ✓ LOO of order 10^{-2} to 10^{-1}
  - ✓ Performance validation

UQpyLab

300 NSCD model runs vs *300 metamodel runs*

Impact test: 1020kJ

Impact test: 520kJ



Rest

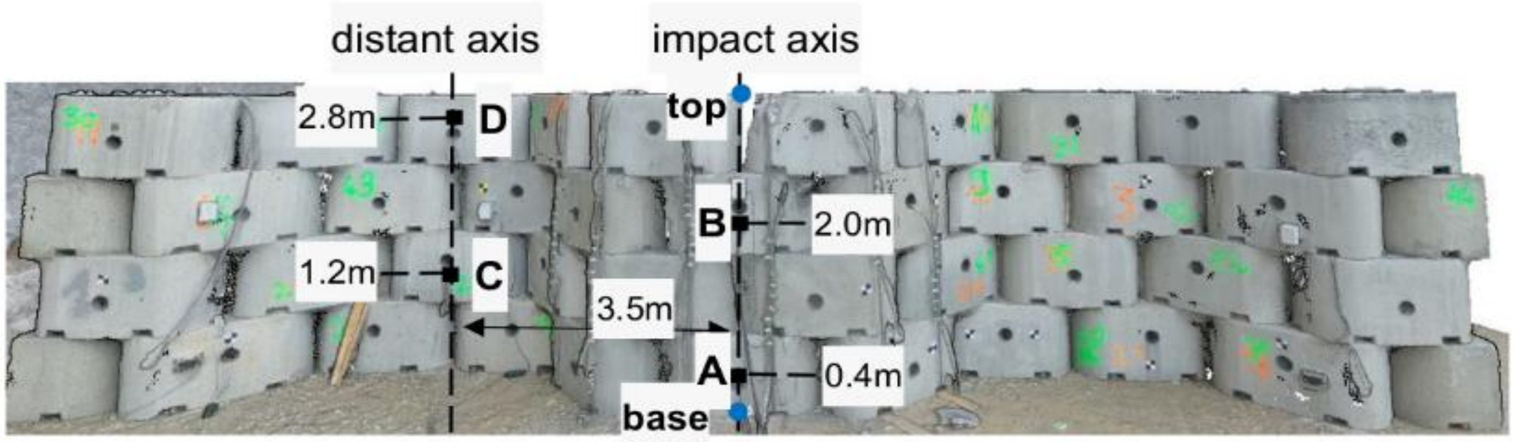
Model calibration

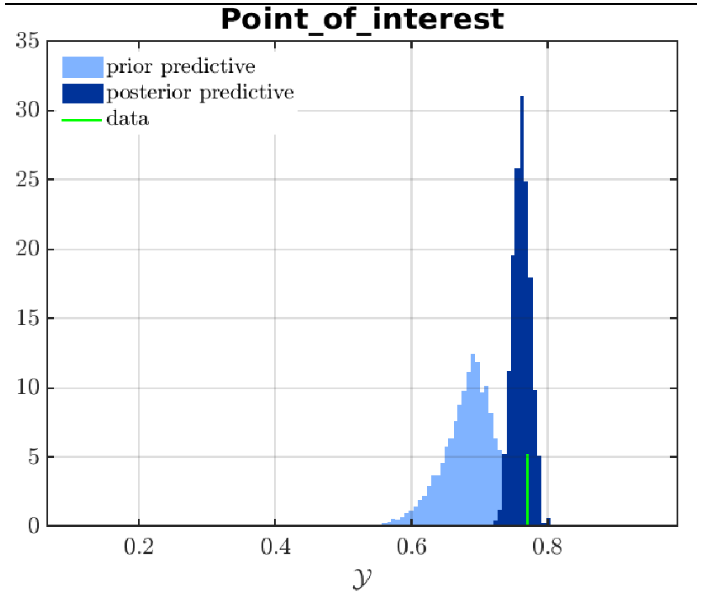
UQLab

Purpose, strategy and success

Bayesian inference to obtain the best set of input parameters

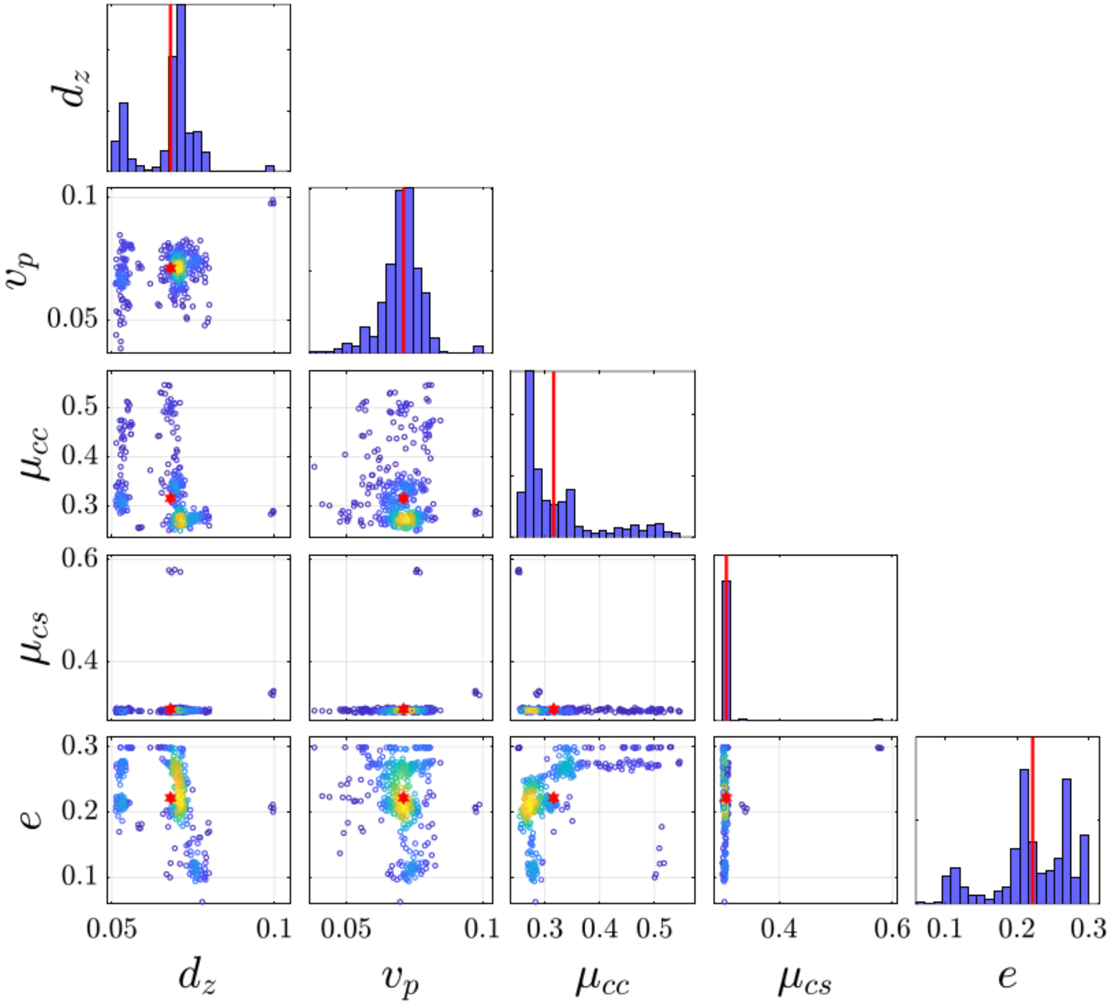
For each impact test	Location	Top			Base			C			D		
	Time	Init	max	rest	Init	max	rest	Init	max	rest	Init	max	rest
Calibration Approach	Instant	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Point	✓			✓			✓			✓		
	Energy	✓											
	All (0.5,1 MJ)	✓✓											

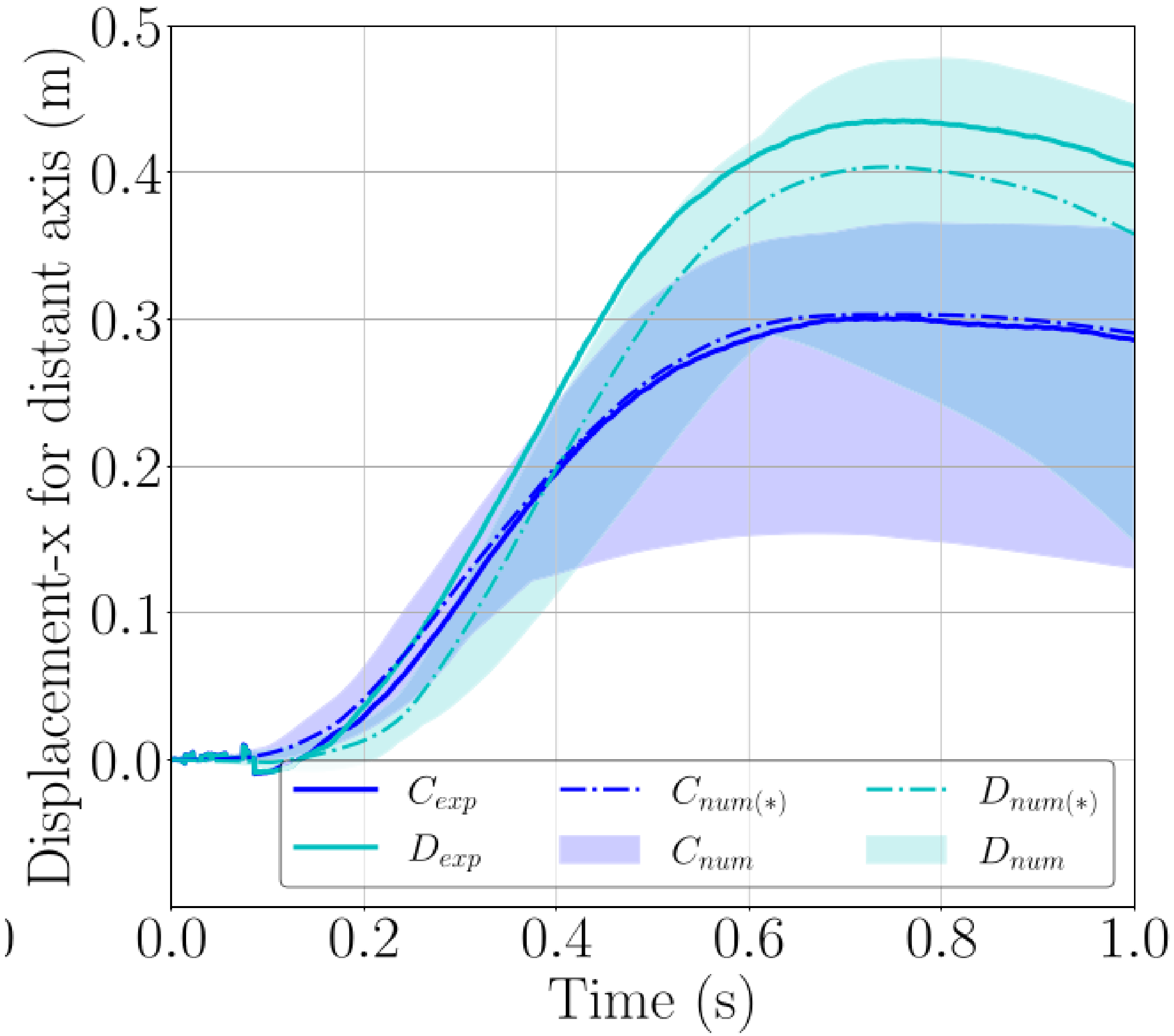




Model calibration

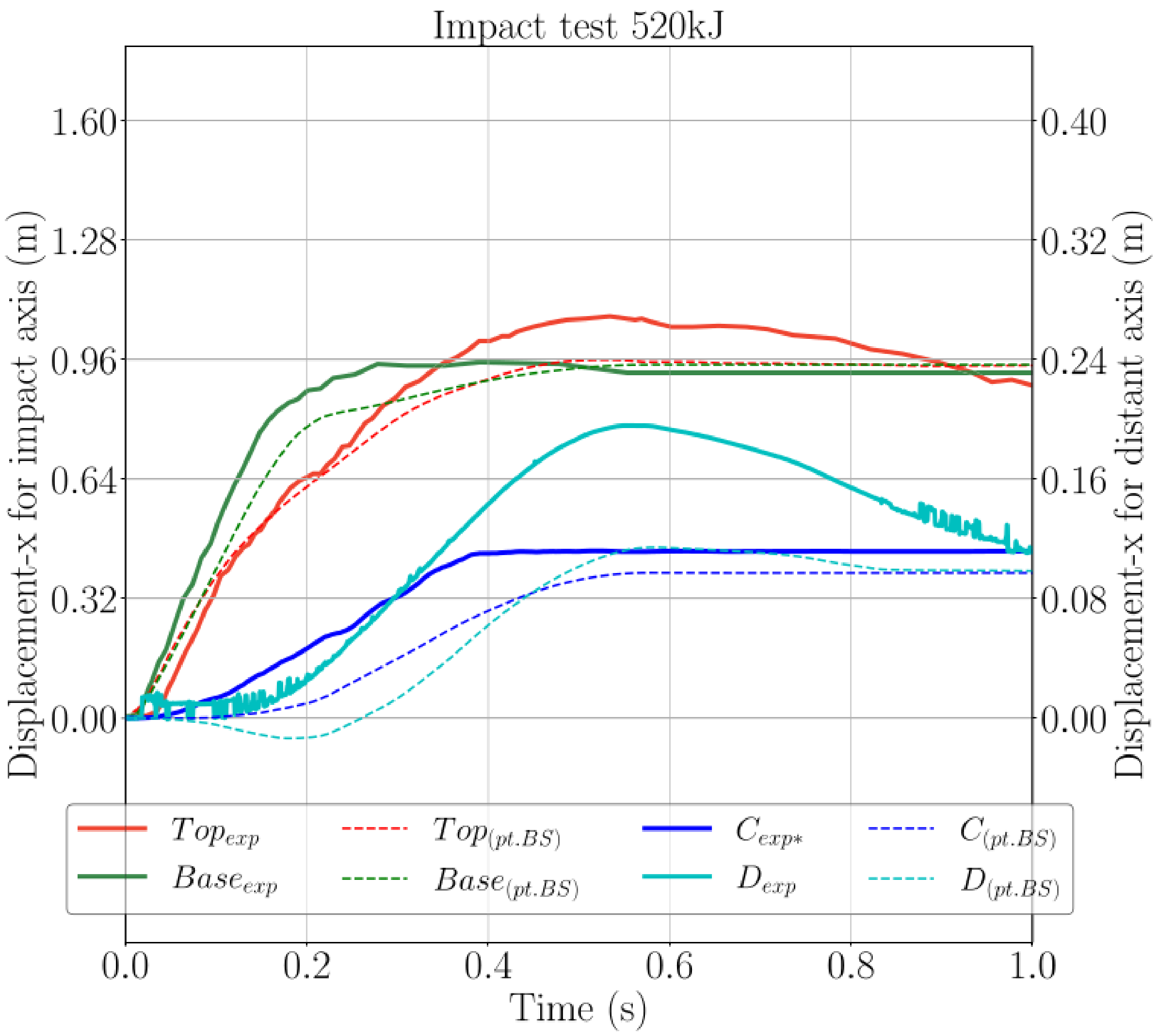
Purpose, strategy and success

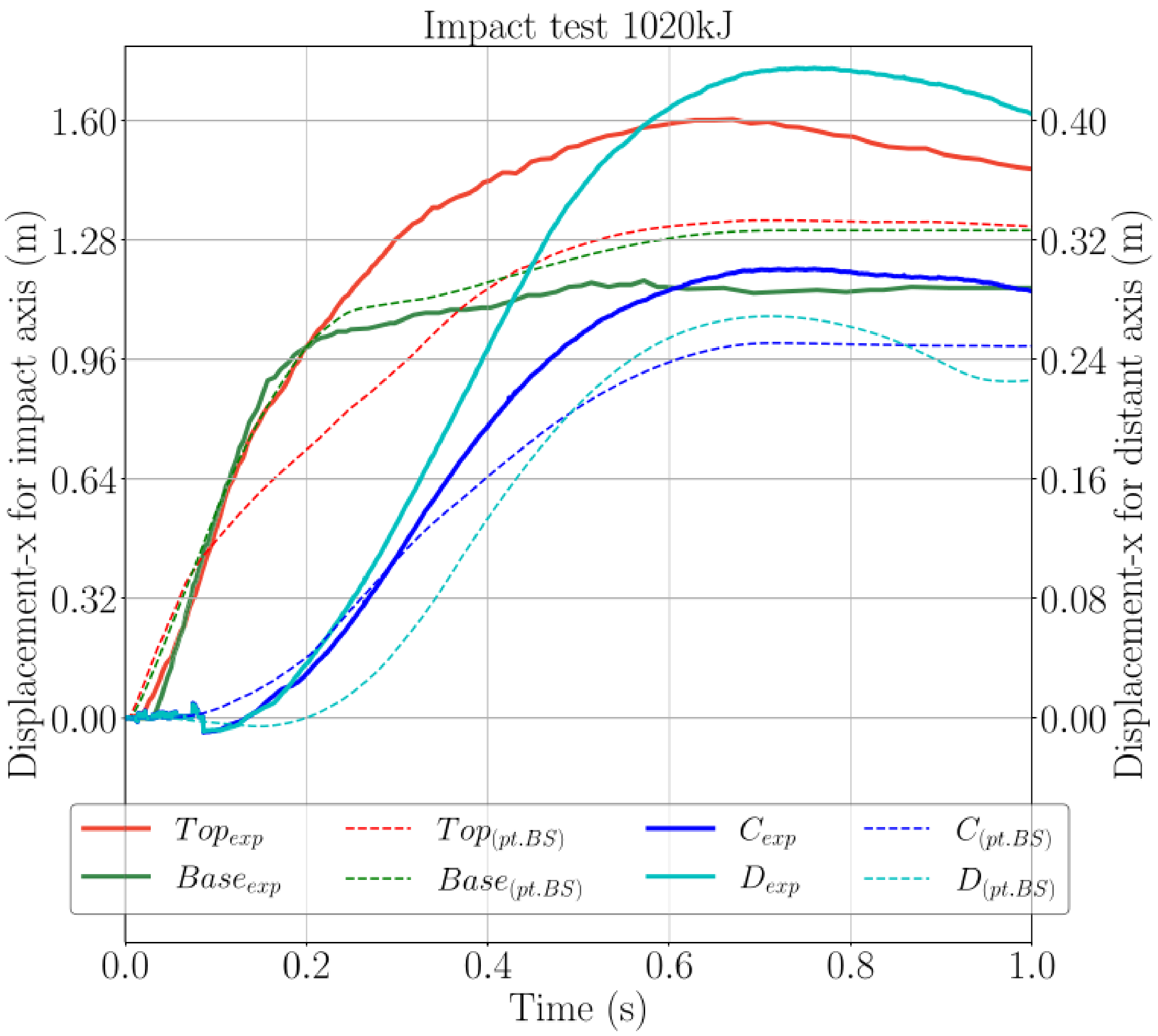




Model calibration

Purpose, strategy and success





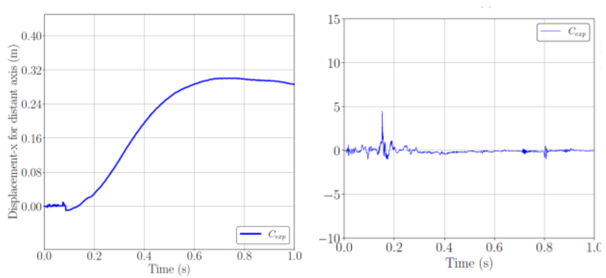
Conclusions (Part 1 – Model calibration)

Let’s recap

- ✓ NSCD model of articulated concrete structure is well-defined in Siconos, inclusive of all on-site features,
- ✓ Mechanical and dynamic response is strategically computed,
- ✓ High confidence in model performance is established through thousands of model runs,
- ✓ On-site measurements are reproduced,

Work continuation…

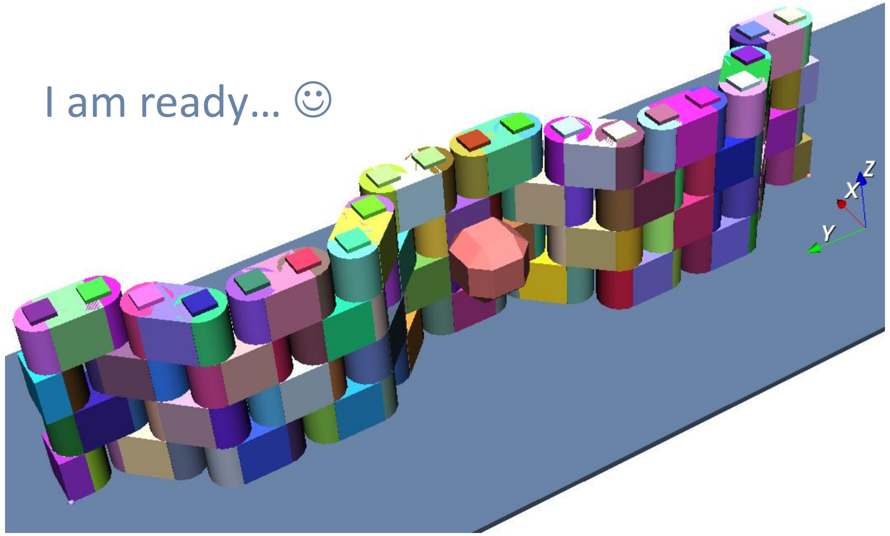
NSCD Model calibrated – milestone achieved … Now, what's next?

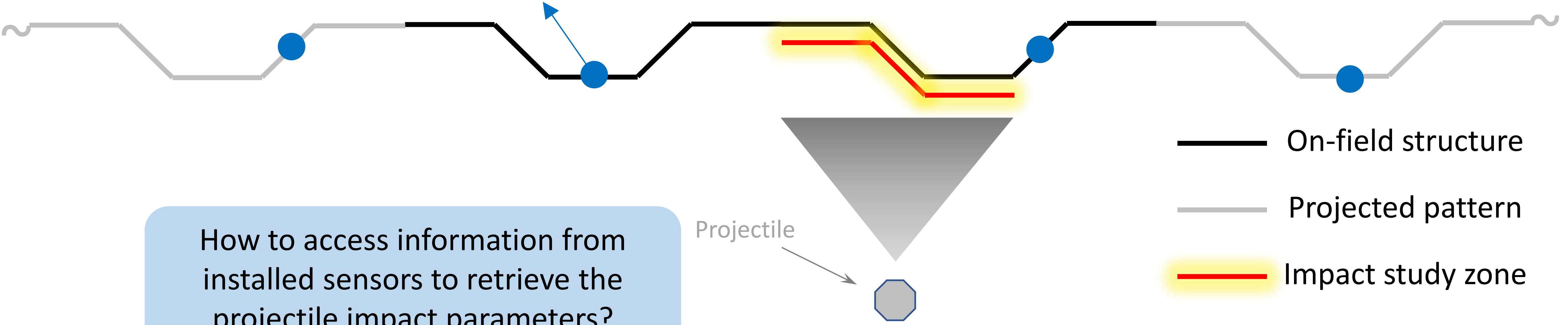


Data acquisition

(displacement/acceleration)

I am ready… ☺

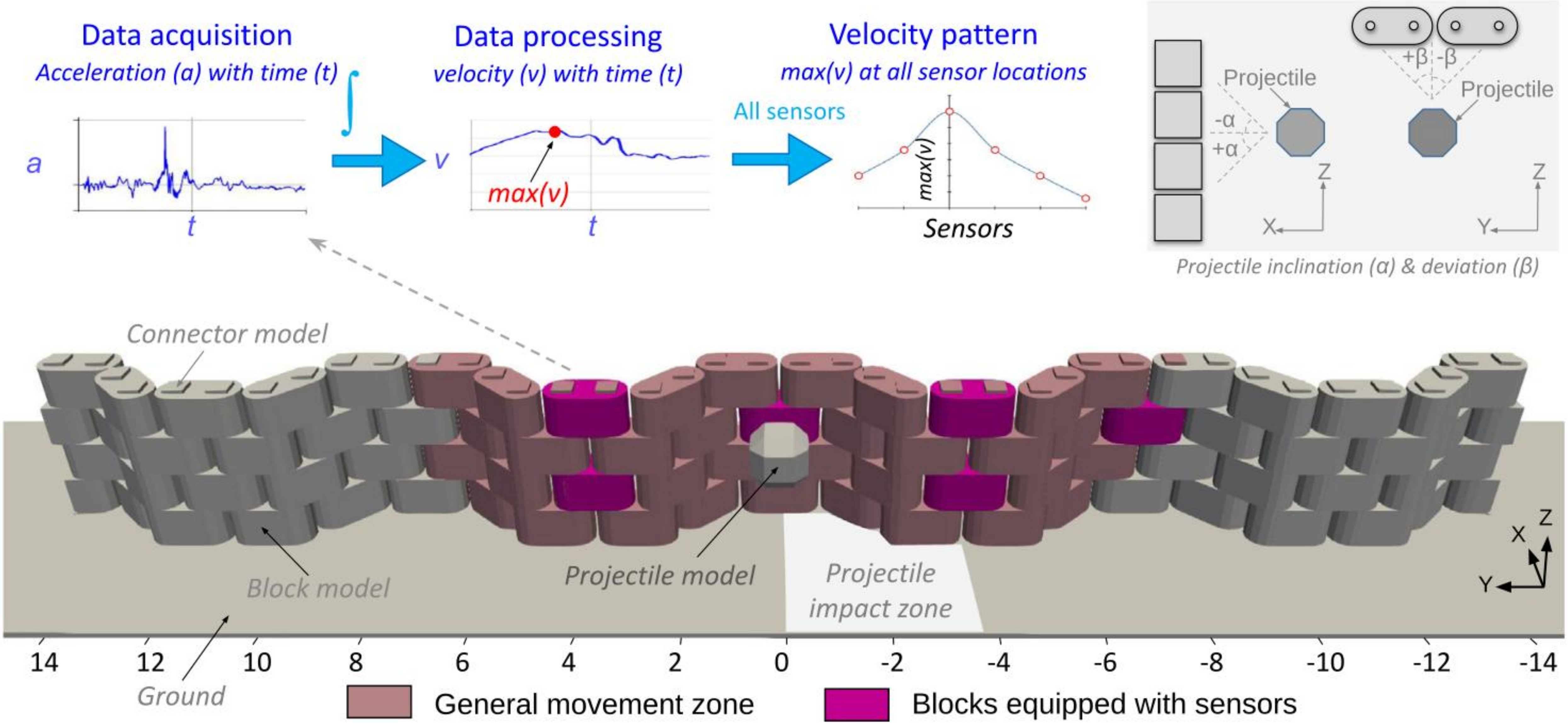




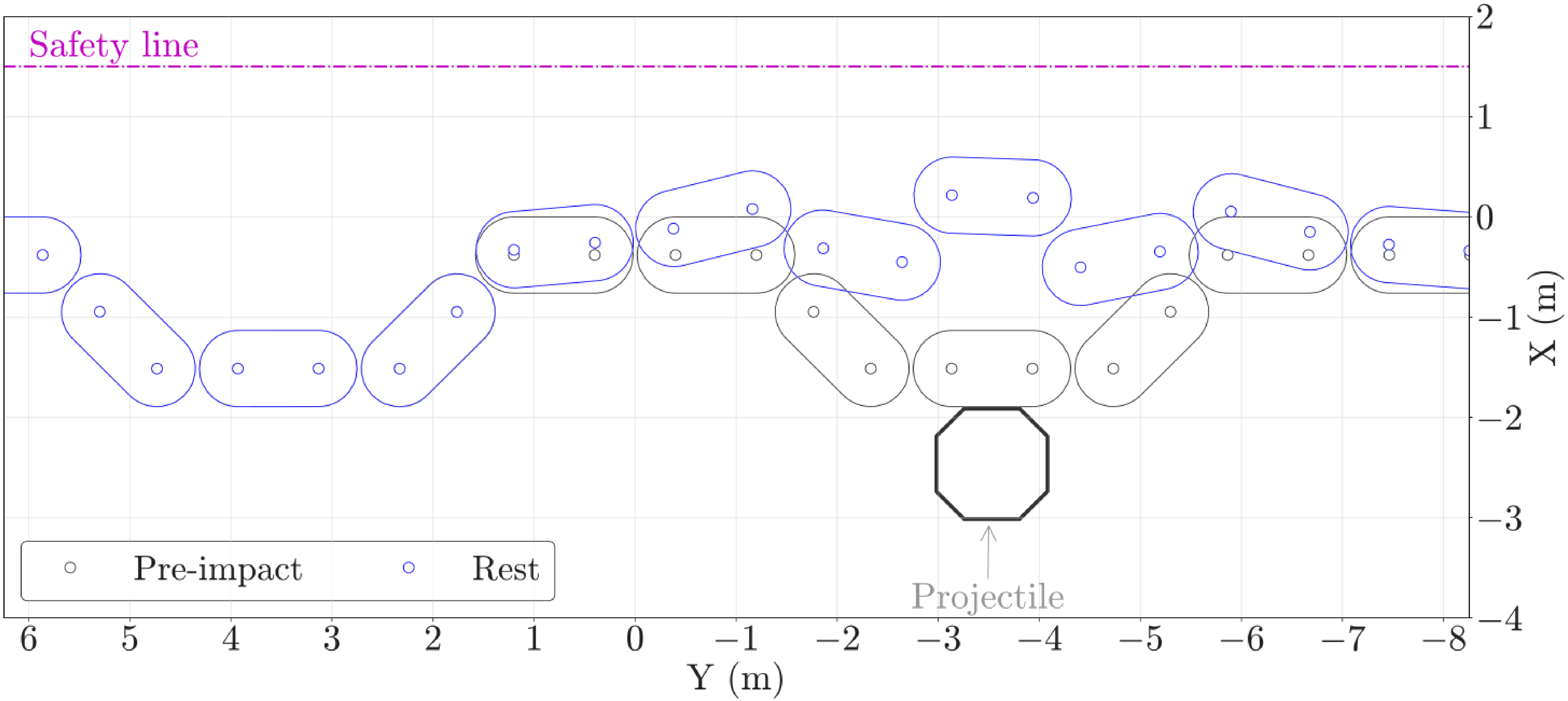
How to access information from installed sensors to retrieve the projectile impact parameters?

The stochastic analysis is guiding our way

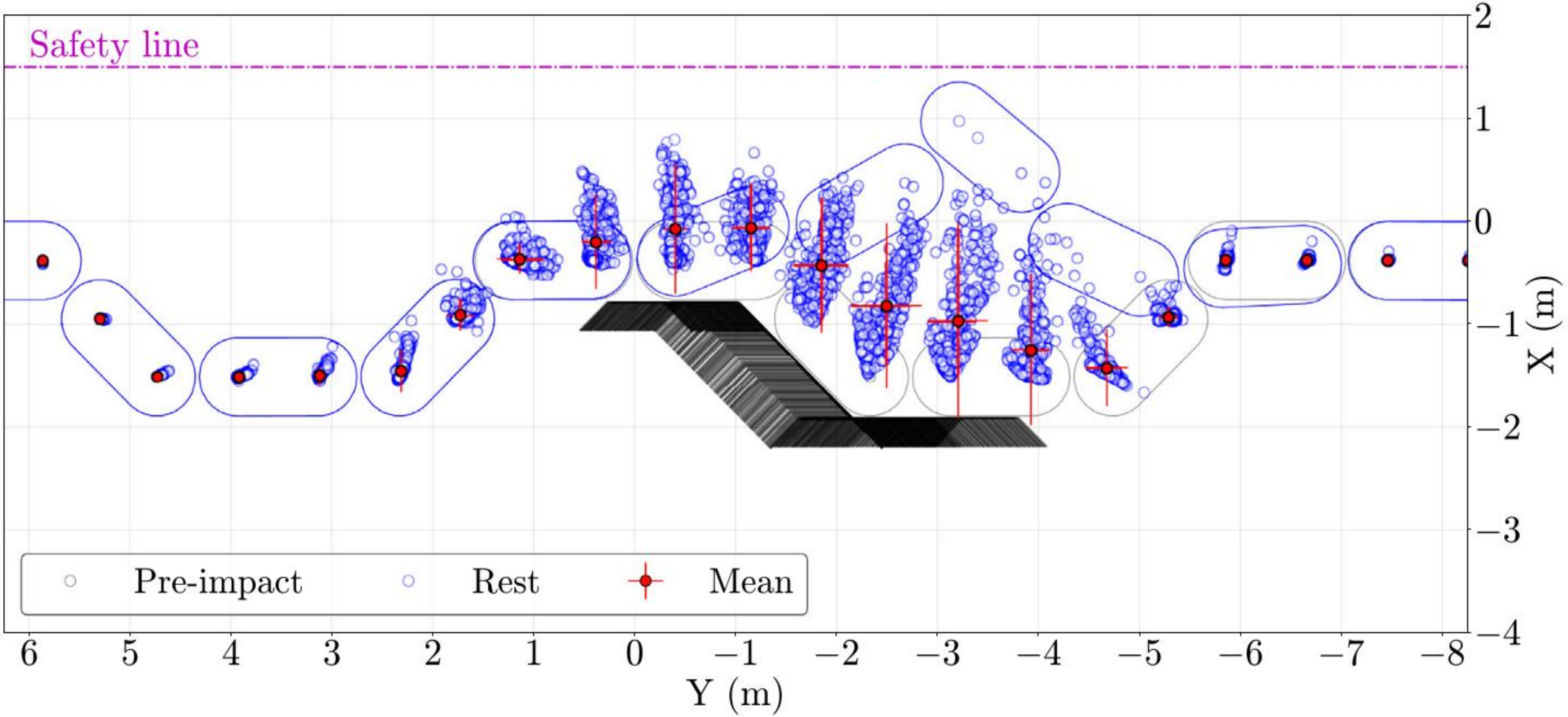
Model inverse analysis



Projectile impact at different position – illustration



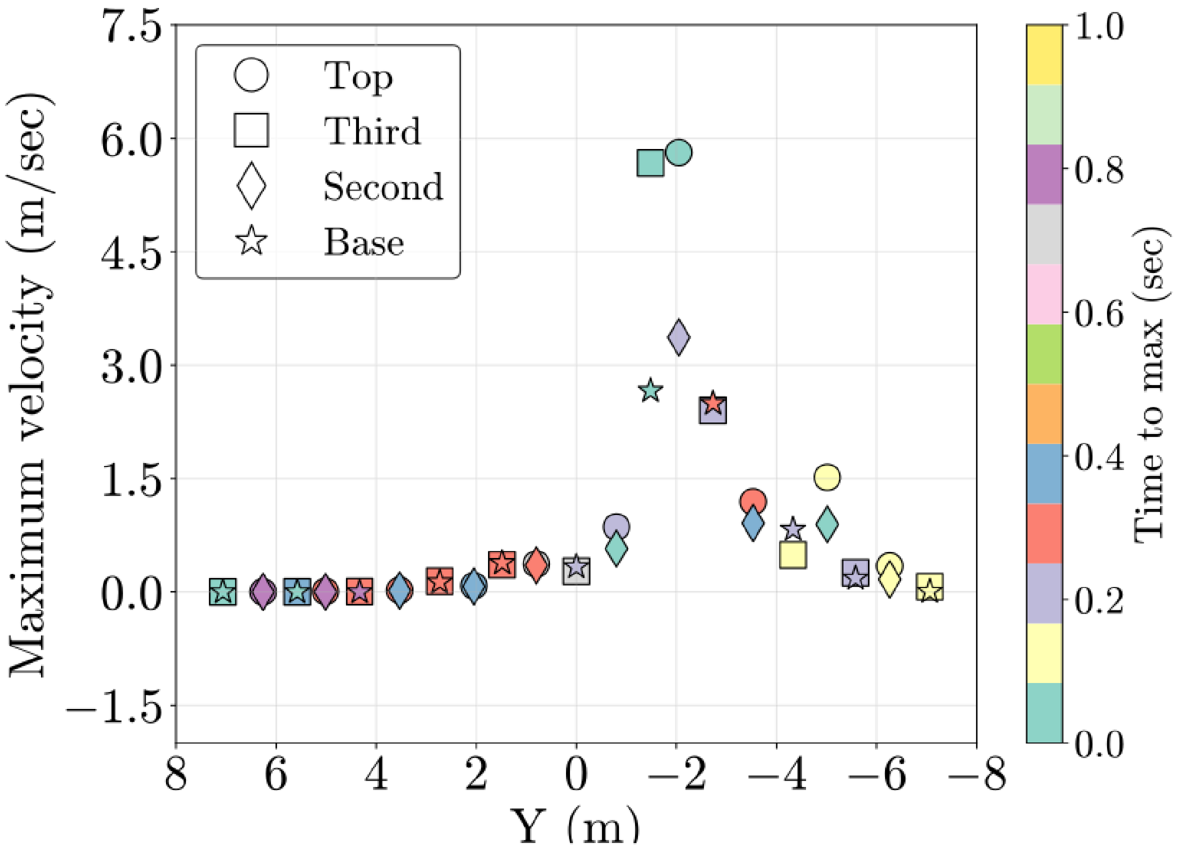
Projectile impact at different location - General

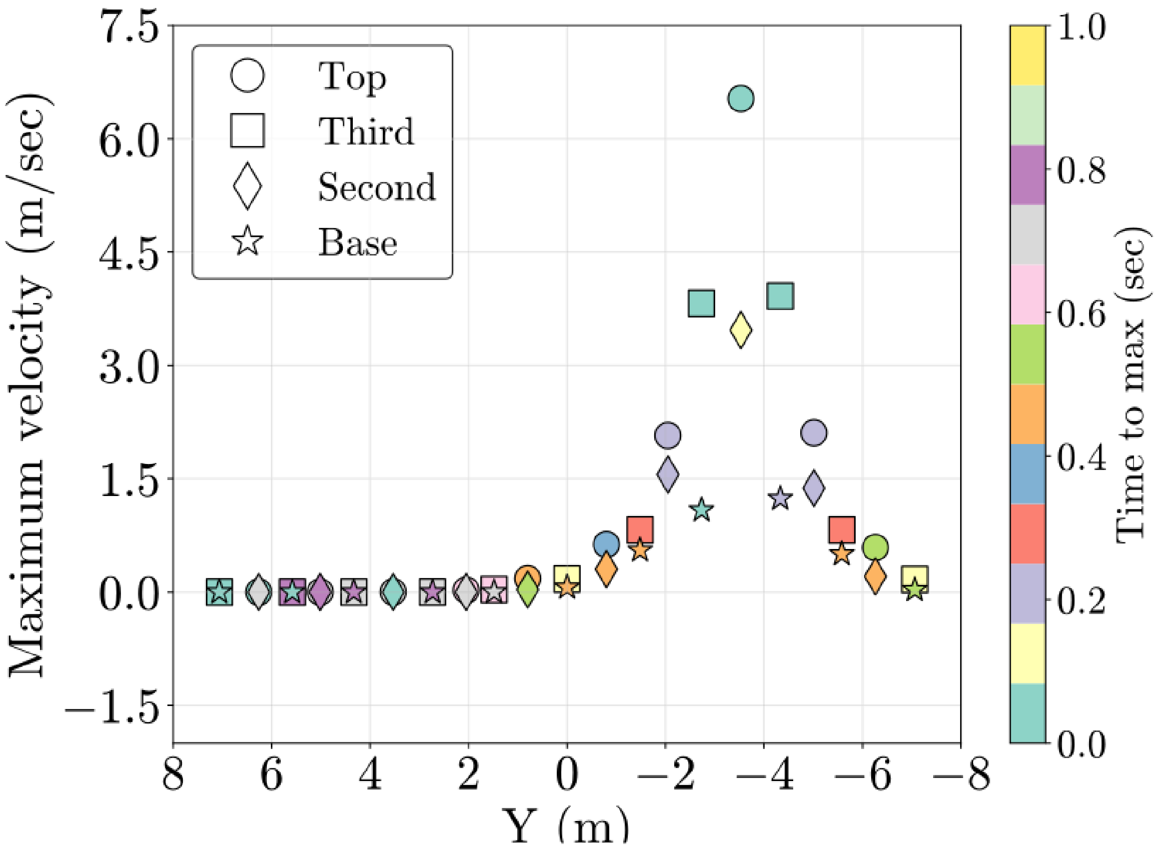


10 20
v
2.5 5.0
Ω
1 2 3
y
1 2
z
−50 0 50
α
−25 0 25
β

Projectile impact at different location - illustration

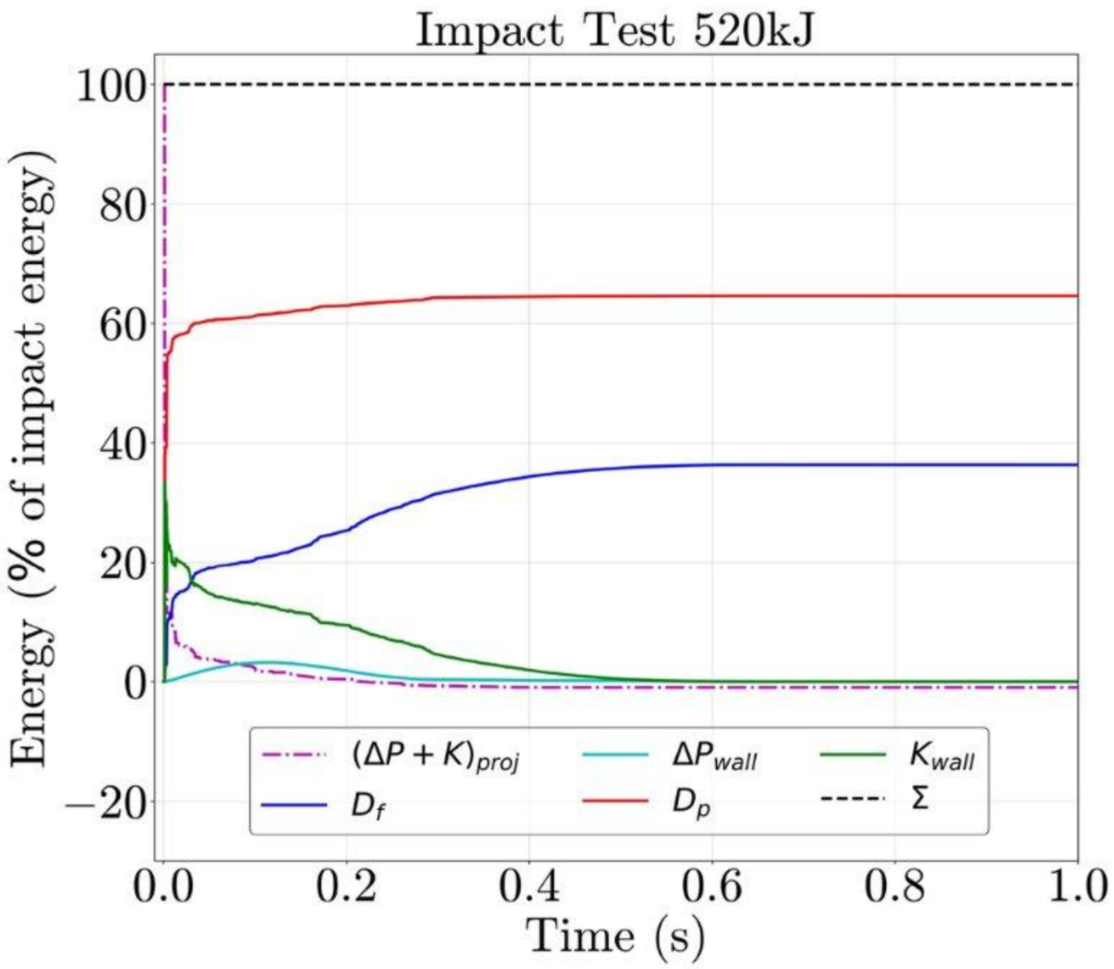
Maximum velocity pattern

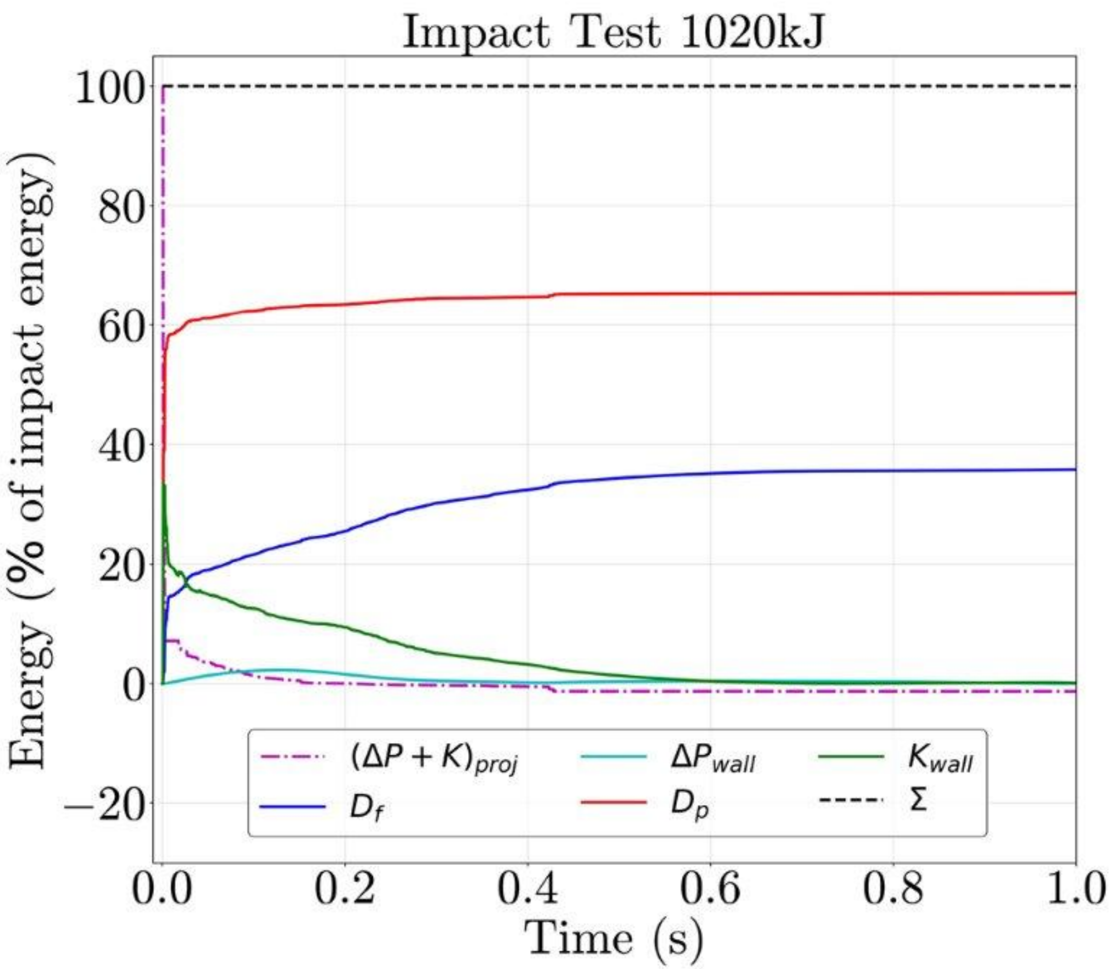




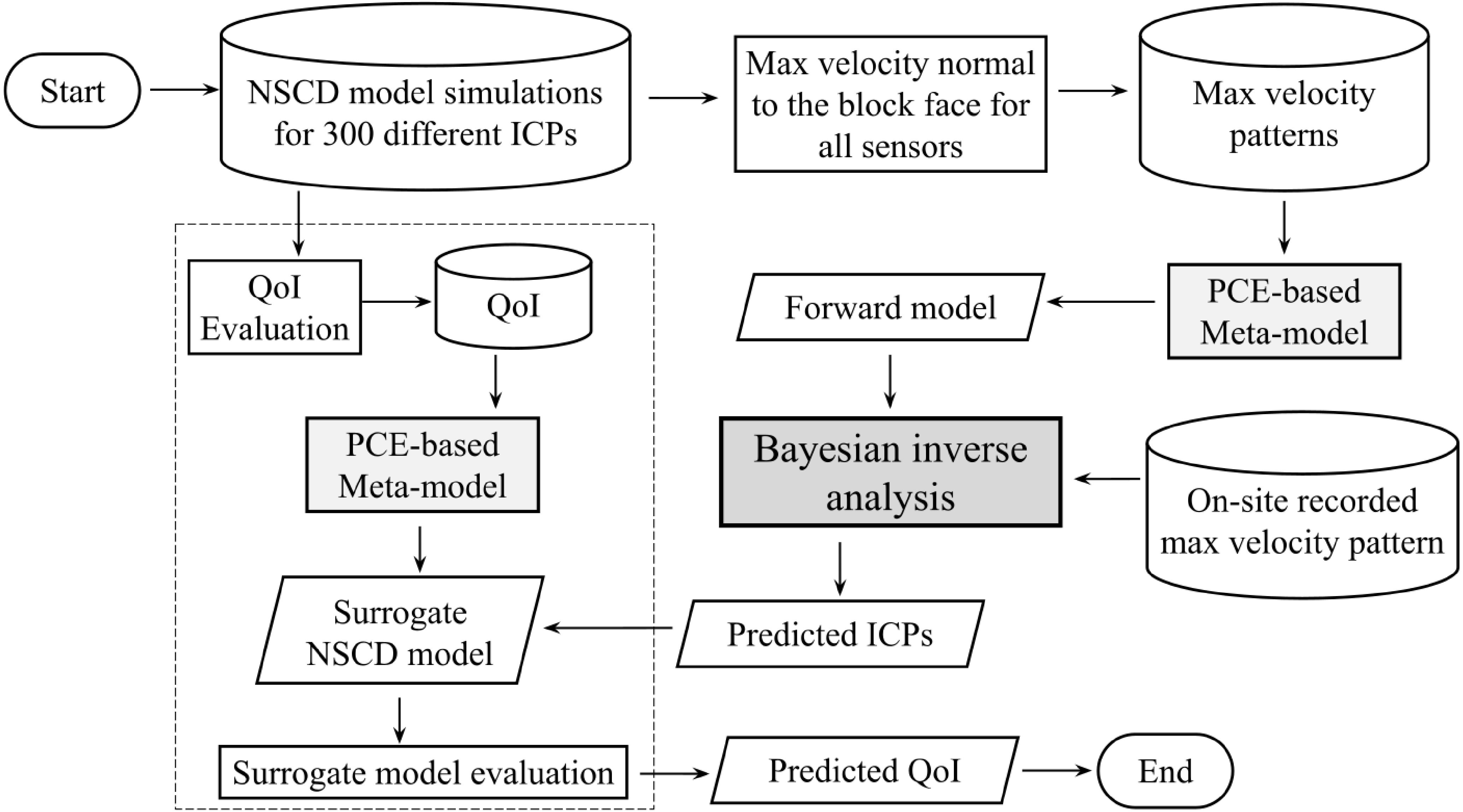
Energy balance computation

Incident energy of projectile = wall movement + plastic damage + friction + remaining projectile energy



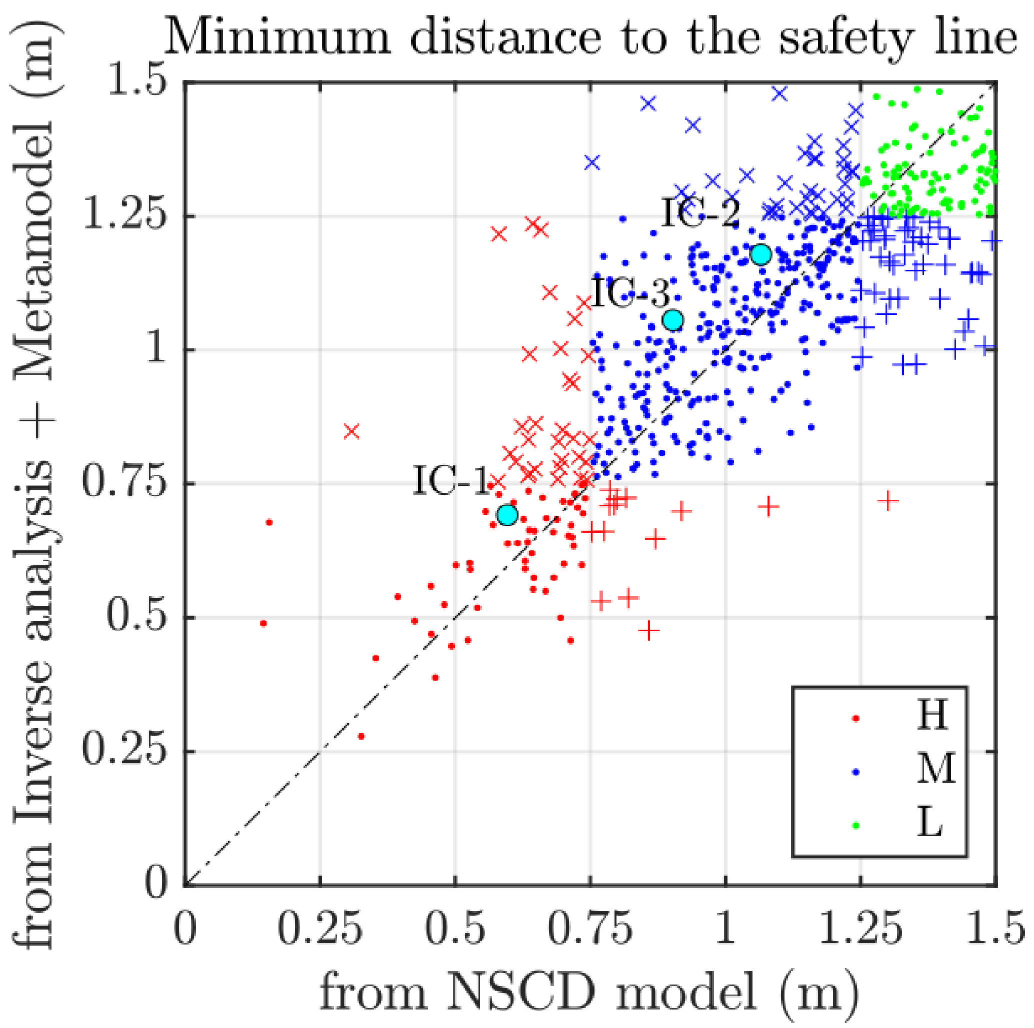


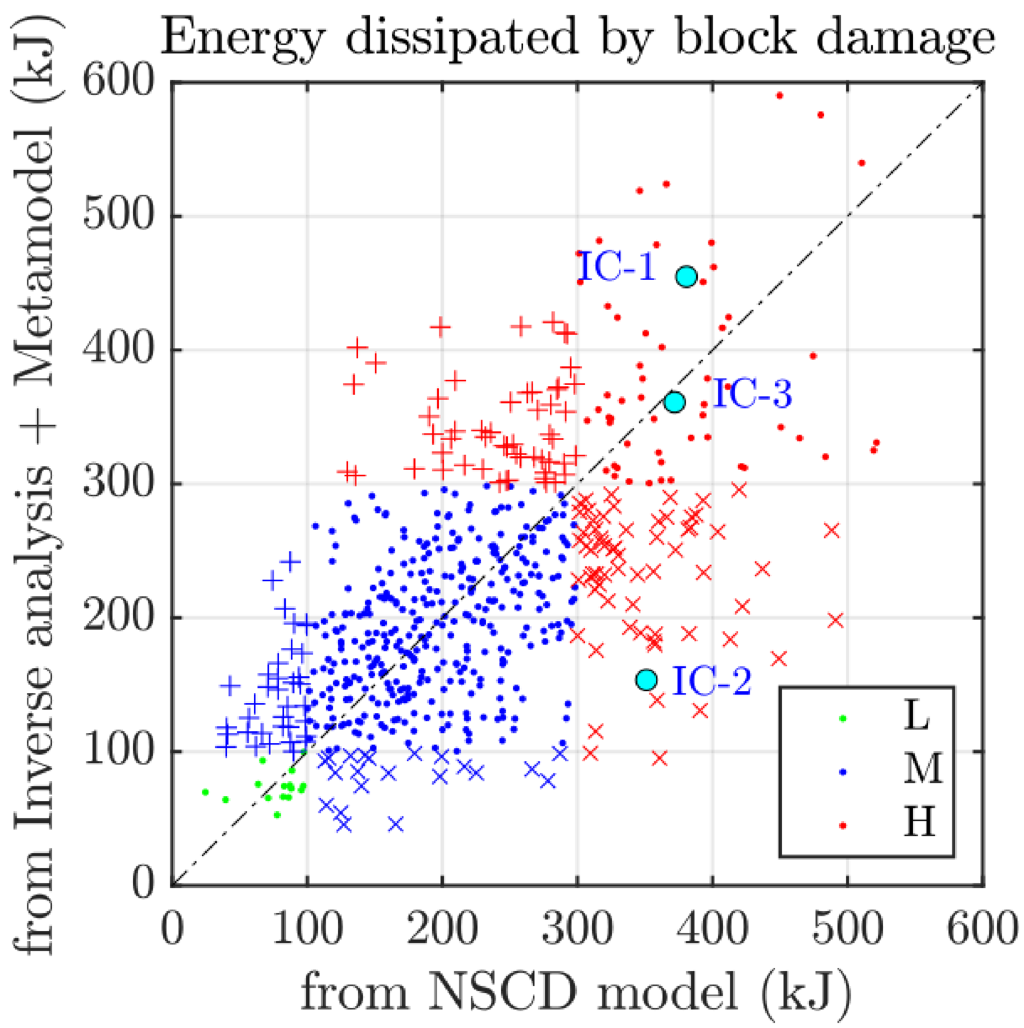
Workflow for inverse analysis



QoI – Impact warning on real-time

Distance to safety line and block damage





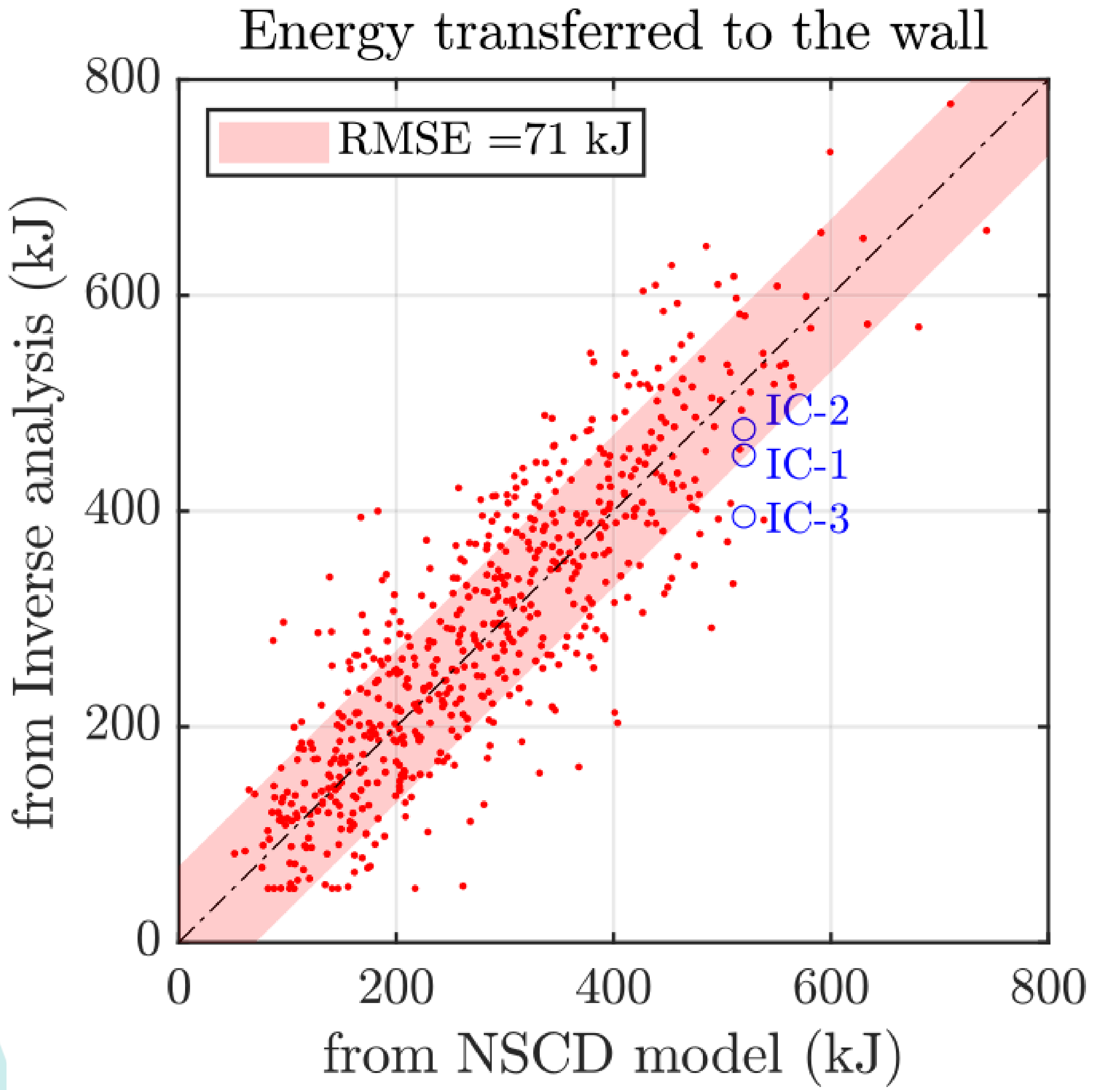
QoI – post-impact assessment

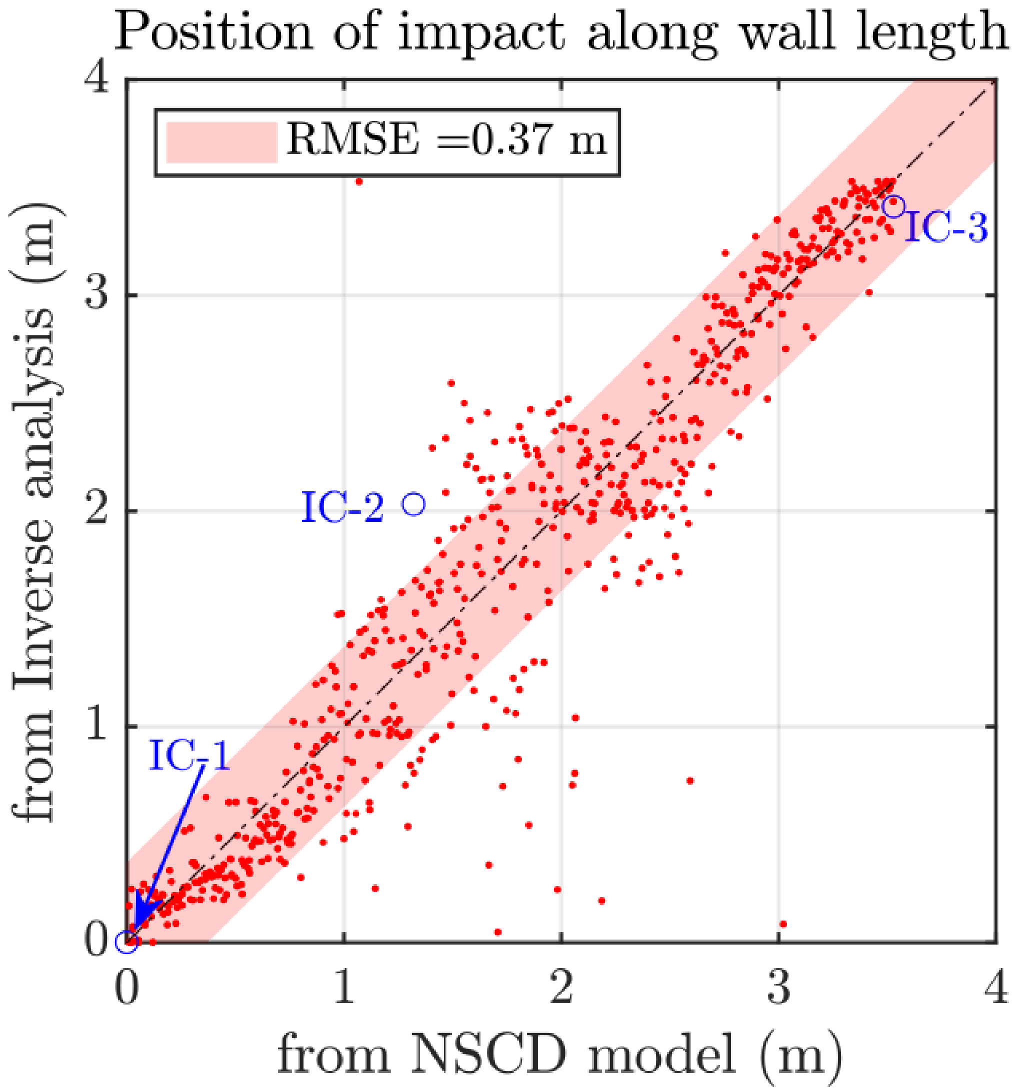
Reduced dimension (from 6 variables) to 4 variables

$$K_{proj} = \frac{1}{2} m \left[v_{xy}\right]^2 + \frac{1}{2} m \left[v_z\right]^2 + \frac{1}{2} I \left[2\pi\Omega\right]^2$$

$$v_z = v \sin\alpha$$

$$v_{xy} = v \cos\alpha$$

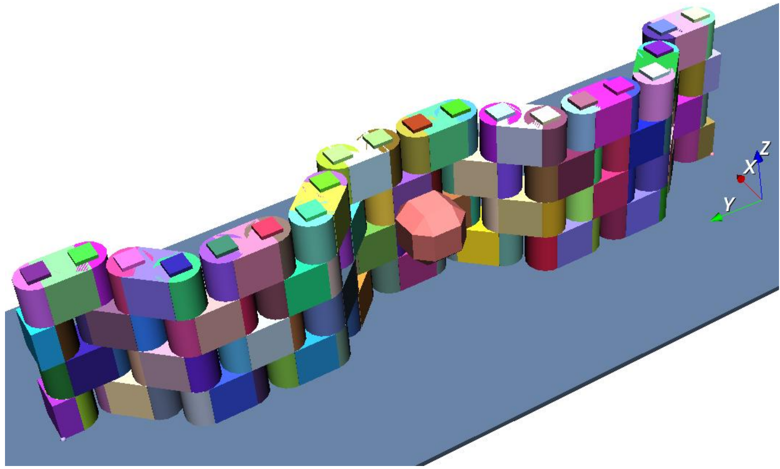




Conclusions (Part 2 – Inverse analysis)

Let's recap

- ✓ Possible to carry out inverse analysis on a complex structure exposed to localized dynamic loading,
- ✓ Knowledge of a site-specific impact condition range is perquisite,
- ✓ The framework is transferable to other protection structures (such as flexible barriers) ,
- ✓ Possible to assess other wall configurations ,



Work publication

- Lambert S., **Gupta R.**, Bourrier F., Acary V. (2024) Energy dissipation in rockfall protection structures quantified from numerical modelling: application to a NSCD model of an articulated wall. *Rock Mechanics and Rock Engineering* - 58, 1957–1973. 10.1007/s00603-024-04222-9
- **Gupta R.**, Bourrier F., Lambert S. (2024). Bayesian inference based inverse analysis of the impact response of a rockfall protection structure: Application towards warning and survey. *Engineering Structures* - 321, 118800. 10.1016/j.engstruct.2024.118800
- **Gupta R.**, Bourrier F., Acary V., Lambert S. (2023). Bayesian interface based calibration of a novel rockfall protection structure modelled in the non-smooth contact dynamics framework. *Engineering Structures*, 297, 116936. 10.1016/j.engstruct.2023.116936

Grateful for your time and attention

Thank you